



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	9 MARCH 2022
GRADE	11	MARKS	50
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	1 HOUR
CLASS			

INSTRUCTIONS:

1. This paper consists of 5 pages. Please check that your question paper is complete.
 2. Answer all questions on the folio paper provided.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. It is in your own interest to write legibly and to present your work neatly.
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QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	9	
2	19	
3	11	
4	11	
TOTAL: 50		

Question 1:

- 1.1. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question on the folio paper provided. **(5)**

1.1.1. Which of the following is not an excretory organ?

- A. Lungs
- B. Skin
- C. Liver
- D. kidney

1.1.2. Which of the following factors could possibly result in a person excreting more urine than normal?

- A. A high reabsorption of water in the proximal tubules
- B. A low concentration of ADH in the blood
- C. Low blood pressure because of too little tissue fluid.
- D. Excessive intake of salty foods.

1.1.3. The walls of blood capillaries in the kidney are perforated (have slits). How does this help in the production of urine?

- A. A larger volume of fluid can be filtered out of the blood per minute.
- B. Small molecules are allowed through but not larger ones.
- C. Higher blood pressures can be tolerated without damage to the capillaries.
- D. Phagocytes are able to escape from the capillaries and prevent kidney infections.

1.1.4. Which of the following receives blood from the efferent arteriole?

- A. The renal vein
- B. The glomerulus
- C. The afferent arteriole
- D. The peritubular capillary network

1.1.5. Which of the following does the renal system NOT play a direct role in?

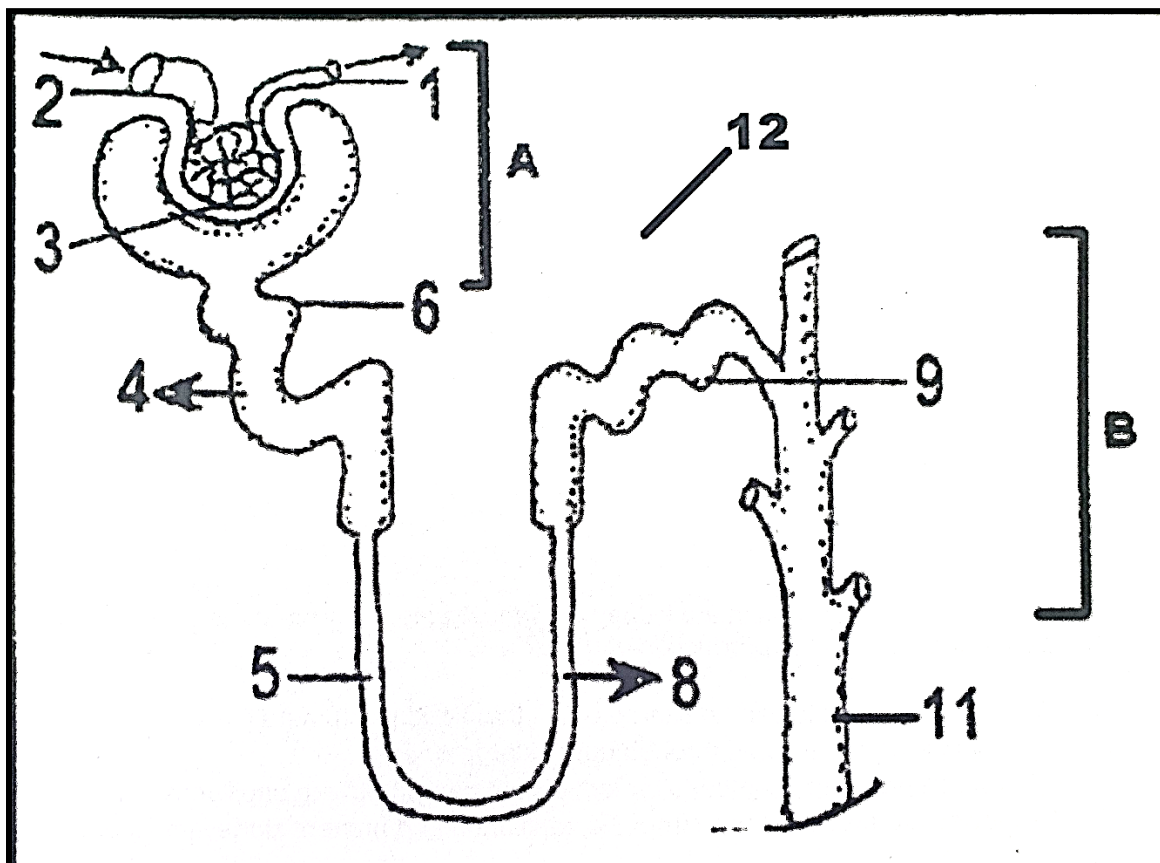
- A. Regulation of blood solute concentrations
- B. Blood temperature regulation
- C. Blood pressure regulation
- D. Blood pH regulation

- 1.2. Indicate whether each of the statements in COLUMN I applies to A only, B only, both A and B or none of the items in COLUMN II. Write A only, B only, both A and B or none next to the question number on your answer sheet. (4)

COLUMN I	COLUMN II
1.2.1. Active reabsorption	A: Glucose B: Uric acid
1.2.2. Nitrogenous waste	A: Urea B: Uric acid
1.2.3. Secreted by the adrenal gland	A: Adrenalin B: ADH
1.2.4. Removal of urine from the bladder	A: Secretion B: Excretion

Question 2:

Study the diagram below and answer the questions that follow.



- 2.1. Provide the name of structure A. (1)

- 2.2. In which areas of the kidney would structures 3 and 5 be found respectively? (2)
- 2.3. Name ONE substance that is indicated by the arrow at label 4 and ONE substance that would be exiting at label 8. (2)
- 2.4. Explain the significance in the structural difference between label 1 and label 2. Provide the names of these structures in your explanation. (3)
- 2.5. Urine formation increases during winter seasons. Provide a reason for this. (3)

An investigation was performed to determine the flow rate and composition of fluids in the human kidney. Samples of the fluid were taken at regions 1, 12, 6 and 11 as shown in the diagram above. The results are recorded in the table below.

Part of nephron	Rate of flow of fluids (cm ³ /min)	Solute concentration (g/cm ³)		
		Albumin (protein)	Amino acids	Uric acids
1	500	4.5	0.2	0.03
2	2000	0.0	0.2	0.03
6	10	0.0	0.1	0.12
11	1	0.0	0.1	1.70

- 2.6. With reference to the table, provide an explanation for each of the following observations:
- 2.6.1. The decrease in the flow rate from region 2 to region 1. (3)
- 2.6.2. The difference in albumin concentration between 1 and 6. (3)
- 2.6.3. The change in uric acid concentration between 6 and 11. (2)
- (19)**

Question 3

- 3.1. Define the term 'excretion'. (2)
- 3.2. If a person's kidneys do not function as they should a person may be put onto dialysis as a short term solution.
- a. What is the term for the loss of proper kidney functioning? (1)
- b. Provide ONE disadvantage of dialysis as a treatment. (1)

- c. What is the long-term treatment for the renal problem mentioned in a? (1)
- d. Mention ONE challenge associated with the long-term treatment. (1)
- 3.3. Draw a labeled diagram of a longitudinal section through the kidney. (5)
- (11)**

Question 4

The table below shows the concentration of ADH in the blood at different plasma solute concentrations.

Plasma solute concentration (mOs/kg)	Plasma concentration of ADH (arbitrary unit)
270	0.0
275	0.0
280	0.0
285	0.0
290	1.0
295	2.0
300	3.0
305	4.0
310	6.0

- 4.1. Define osmoregulation. (1)
- 4.2. Plot the information above in an appropriate graph. (5)
- 4.3. At which solute concentration would ADH be released? (1)
- 4.4. Which structure is responsible for the secretion of ADH? (1)
- 4.5. Define the role ADH if there is too little water in the body. [A flow diagram may be used.] (3)
- (11)**

TOTAL: 50 MARKS